
ME 303P : Heat Transfer Lab

Course Code : ME 303P

Course Name : Heat Transfer Lab

L-T-P-C : 0-0-2-1

Intended for : UG

Prerequisite : ME 303 - Heat Transfer & ME - 213 Engineering Thermodynamics, ME - 210 Fluid Mechanics

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- NA

Laboratory/practical/tutorial Modules:

1. Calorific Value Measurement
2. Critical Heat Flux Measurement
3. Heat Transfer from Pin Fin Measurement
4. Heat Transfer in Natural Convection
5. Parallel Flow and Counter Flow Heat Exchanger
6. Radiation Law Verification
7. Refrigeration Cycle Verification
8. Air-Conditioning Cycle Verification - Open Cycle
9. Air-Conditioning Cycle Verification - Closed Cycle
10. Two-Phase Heat Transfer
11. Lumped Heat Transfer Analysis
12. Unsteady Heat Transfer Analysis
13. Analysis of Water Cooler

Textbooks:

1. J.P. Holman, **Experimental Methods for Engineers**, 7th edition, Tata McGraw-Hill 2001.
2. T.G. Beckwith, J.H. Lienhard V, R. D. Mamgani, **Mechanical Measurements**, 5th edition, Pearson Education, 2010.
3. E.O. Doebelin, **Measurement systems, Application and Design**, 5th edition, Tata McGraw-Hill, 2008
4. Incropera and Dewitt, **Fundamentals of Heat and Mass Transfer**, Wiley India.



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5. Cengel, **Heat and Mass Transfer**, Tata McGraw Hill.
 6. Krieth and Bohn, **Principles of Heat Transfer**, Cengage Learnings.

References:

1. NA